

DIGITAL LOGIC AND COMPUTER DESIGN

TEAM DLCD

Department of CSE and CSE (AI & ML)

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Sequential Building Blocks

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Exercise problems: 5.43

Exercise 5.43: Build a 32-bit synchronous Up/Down counter. The inputs are Reset and Up. When Reset is 1, the outputs are all 0. Otherwise, when Up = 1, the circuit counts up, and when Up = 0, the circuit counts down.

Given

- Inputs: Reset, Up, CLK
- Output: Q[31:0]
- If Reset = 1 → Q = 0
- If Reset = 0 and Up = 1 → count **up**
- If Reset = 0 and Up = 0 → count **down**

Design

Use:

- **32-bit register**
- **32-bit adder/subtractor**
- **MUX**
- Constant 1

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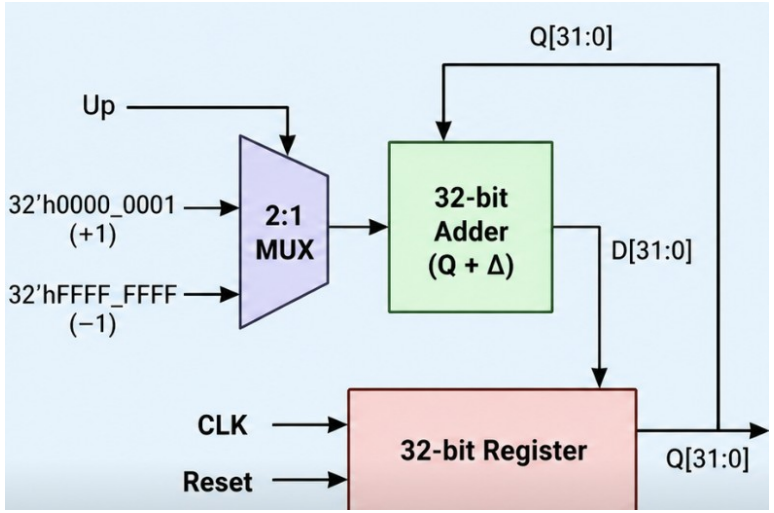
Exercise problems: 5.43

The next-state equation is:

Q+= { 0 Reset=1
Q+1 Reset=0, Up=1
Q−1 Reset=0, Up=0}

A convenient implementation is: Q^(+) = Q + (2Up-1)

or, more practically, use an adder/subtractor controlled by Up.



Reset here is synchronous - it overrides D on the clock edge, not an immediate async clear.

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Exercise problems: 5.44

Exercise 5.44 Design a 32-bit counter that adds 4 at each clock edge. The counter has reset and clock inputs. Upon reset, the counter output is all 0.

Given

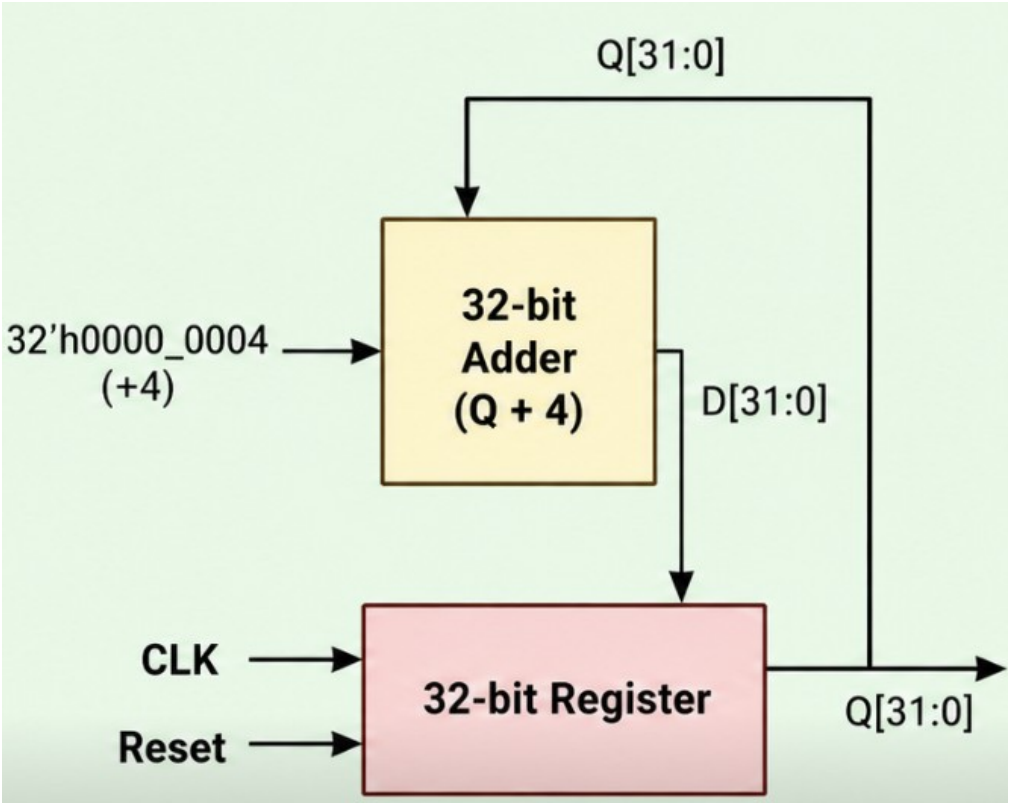
- Inputs: Reset, CLK
- Output: Q[31:0]
- Reset → Q = 0
- Every clock edge:

$$Q^{(+)} = Q+4$$

Design

Use:

- **32-bit register**
- **32-bit adder**
- Constant 4



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Exercise problems: 5.51

Implement the following functions using a single 16 × 3 ROM. Use dot notation to indicate the ROM contents.

(a) $X = AB + B\overline{C}D + \overline{A}\,\overline{B}$

(b) $Y = AB + BD$

(c) $Z = A + B + C + D$

Given:

(a) $X = AB + B\overline{C}D + \overline{A}\,\overline{B}$

(b) $Y = AB + BD$

(c) $Z = A + B + C + D$

A **16 × 3 ROM** has:

- **4 address inputs:** A,B,C,D since 2^(4)=16 addresses
- **3 data outputs:** X,Y,Z

Step 1: Determine the ROM contents

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Exercise problems: 5.51

Address	A B C D	X	Y	Z	ROM Data (XYZ)
0	0000	1	0	0	**100**
1	0001	1	0	1	**101**
2	0010	1	0	1	**101**
3	0011	1	0	1	**101**
4	0100	0	0	1	**001**
5	0101	0	1	1	**011**
6	0110	0	0	1	**001**
7	0111	0	1	1	**011**
8	1000	0	0	1	**001**
9	1001	0	0	1	**001**
10	1010	1	0	1	**101**

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Exercise problems: 5.51

11	1011	0	0	1	**001**
12	1100	1	1	1	**111**
13	1101	1	1	1	**111**
14	1110	1	1	1	**111**
15	1111	1	1	1	**111**

Step 2: ROM contents

Therefore, the **16 × 3 ROM** stores:

100, 101, 101, 101, 001, 011, 001, 011, 001, 001, 101, 001, 111, 111, 111, 111

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Exercise problems: 5.51

Dot Notation

In **dot notation**, place a dot at the intersection of an address row and output column wherever the ROM stores a **1**.

Address ABCDABCD	X	Y	Z
0000	•		
0001	•		•
0010	•		•
0011	•		•
0100			•
0101		•	•
0110			•
0111		•	•

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Exercise problems: 5.51

1000			•
1001			•
1010	•		•
1011			•
1100	•	•	•
1101	•	•	•
1110	•	•	•
1111	•	•	•

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Exercise problems: 6.3

Exercise 6.3: Consider memory storage of a 32-bit word stored at memory word 42 in a byte-addressable memory. (a) What is the byte address of memory word 42? (b) What are the byte addresses that memory word 42 spans? (c) Draw the number 0xFF223344 stored at word 42 in both big-endian and little-endian machines. Your drawing should be similar to Figure 6.4. Clearly label the byte address corresponding to each data byte value.

Given

- Memory word = **42**
- Word size = **32 bits = 4 bytes**
- Memory is **byte-addressable**
- Data = 0xFF223344

(a) Byte address of memory word 42

Since each word occupies **4 bytes**: Byte address=Word number×4=42×4=168

Convert 168 to hexadecimal: 0xA8

Answer:0xA8. So, **word 42 starts at byte address 0xA8**.

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Exercise problems: 6.3

(b) Byte addresses that word 42 spans

A 32-bit word contains **4 bytes**.

Starting at 0xA8, the four bytes occupy consecutive addresses:

Byte	Address
Byte 0	0xA8
Byte 1	0xA9
Byte 2	0xAA
Byte 3	0xAB

Answer: 0xA8, 0xA9, 0xAA, 0xAB

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Exercise problems: 6.3

(c) Store 0xFF223344

First divide the 32-bit number into its four bytes: FF 22 33 44, where:

- FF = Most Significant Byte (MSB)
- 44 = Least Significant Byte (LSB)

The textbook explains that in **big-endian**, the MSB is stored at the lower byte address, while in **little-endian**, the LSB is stored at the lower byte address.

Big-Endian

MSB goes to the lowest address. Therefore:

- FF → 0xA8
- 22 → 0xA9
- 33 → 0xAA
- 44 → 0xAB

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Exercise problems: 6.3

Little-Endian

LSB goes to the lowest address.

Therefore:

- 44 → 0xA8
- 33 → 0xA9
- 22 → 0xAA
- FF → 0xAB

Finally,

Big-endian	**A8:FF, A9:22, AA:33, AB:44**
Little-endian	**A8:44, A9:33, AA:22, AB:FF**

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Exercise problems: 6.4

Exercise 6.4: Repeat Exercise 6.3 for memory storage of a 32-bit word stored at memory word 15 in a byte-addressable memory.

(a) Byte address of memory word 15

Byte Address=15×4=60= 0x3C (in hexadecimal)

(b) Byte addresses that memory word 15 spans

A 32-bit word = **4 bytes**, so it spans:

0x3C, 0x3D, 0x3E, 0x3F

(c) Store 0xFF223344

Endian Type	Byte Address : Data
Big-endian	3C:FF, 3D:22, 3E:33, 3F:44
Little-endian	3C:44, 3D:33, 3E:22, 3F:FF

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THANK YOU

Team DLCD

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